

RUDRA CHAVDA

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EDUCATION

University of California, Santa Barbara

Bachelor of Science, Computer Engineering

Expected June 2027

GPA: 3.95

- **Honors:** 5x Dean's List
- **Coursework:** Data Structures and Algorithms, C Programming, Computer Architecture, Operating Systems, Software Engineering.

TECHNICAL SKILLS

Languages	Python, C, C++, Java, TypeScript, JavaScript, SQL, Assembly
Frameworks	Vue.js, React, Next.js, Node.js, Py4Web, Tailwind CSS, Angular
Tools	Git, Nautilus (Cluster), CARLA, Docker, GDAL, VS Code, Google Colab
Concepts	Reinforcement Learning, Data Structures, Algorithms, System Design

EXPERIENCE

AIEA Lab (UCSC)

Dec 2025 – Present

Undergraduate Researcher (*Robustifying Autonomous Vehicles*)

- Benchmarking Reinforcement Learning (RL) algorithms using **Stable Baselines** to evaluate autonomous vehicle safety.
- Deploying the **CARLA** simulator in a headless environment on the **Nautilus** hyper-cluster (Kubernetes).
- Analyzing edge-case failures by visualizing agent behaviors and reward functions post-training.

CA-Futures (UCSC EEB & Baskin Engineering)

May 2025 – Sep 2025

Software Developer

- Developed a full-stack conservation modeling application using **Vue.js** (frontend) and **Py4Web** (backend).
- Integrated geospatial analysis tools (**GDAL**, Ecoscape) to track species connectivity and visualize ecological data.
- Optimized simulation parameters to process large-scale biological datasets for the Ecology Lab.

Opennote (YC S25)

Jul 2025 – Present

Campus Growth Lead

- Spearheaded campus growth strategy, increasing user engagement by 30% and acquiring 50+ active users.

Fassur

June 2023 – Oct 2023

Front-End Engineering Intern

- Architected a secure, interactive web application using **React.js** and **Tailwind CSS**.
- Designed a responsive front-end framework to improve data visualization for client dashboards.

PROJECTS

Better Notes | *TypeScript, Tailwind CSS, Vercel*

Spring 2025

- Engineered a full-stack note-taking platform with a custom markdown rendering engine for student study workflows.
- Secured 100+ active users within 3 months of deployment to Vercel.
- Live Demo: betternotes-two.vercel.app

Div Explorer Project | *Machine Learning, Python*

Fall 2025

- Developed actionable data labeling pipelines to detect duplicate discoveries and high-overlap patterns.
- Implemented categorization metrics (interpretability, coverage, anomalies) to improve ML model fairness.

Personal Portfolio | *Next.js, Three.js, GSAP*

Winter 2024

- Built a high-performance portfolio site featuring 3D interactive elements using Three.js and GSAP animations.

LEADERSHIP

Eagle Scout Project

Summer 2024

Project Lead, Troop 537

- Directed a team of 20+ Scouts in the construction of redwood structures for Sycamore Grove Park (Livermore).